

The Gravitational Coherence Surface

Registered Ablation Parity in Relational Gravitational Scenes

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Abstract

A gravitational coherence surface (GCS) is a contribution–context parity surface: the locus on which a declared gravitational contribution reaches equality with the context derived from the scene in which it is embedded. The object is geometric and diagnostic. It is not a causal horizon, trapping horizon, force boundary, material membrane, orbital-stability limit, or new source of gravity. Matter, light, and test bodies cross it freely.

The founding formulation expressed contribution and context through declared scalar readability channels. That grammar remains useful in weak-field and measurement-aware settings, but nonlinear general relativity does not ordinarily supply a unique source-by-source decomposition of one fused geometry. The preferred construction is therefore intervention-based. A full scene is solved, a declared finite source is ablated, the comparison scene is solved again, and the two tensor fields are brought onto a common support by an explicit point and frame registration. The primary signature is the registered change in the electric part of the Weyl tensor. The Frobenius amplitude, evaluated under a declared reference inner product, is adopted as the default scalarization for parity extraction, while the registered tensor field remains the principal evidence object.

The ablation response is not interpreted as a privately owned field of the removed source. It is the response of the whole solved scene to that source’s presence, including changes in nonlinear couplings. The GCS is one level set within this generally global relational signature; it does not mark where the signature vanishes. Weak-field acceleration and multipole examples remain useful as declared proxy lanes. Persistent finite-source gravitational identity is retained only as a hypothesis concerning invariants that survive changes of context, frame, scale, representation, and time.

Keywords: general relativity; electric Weyl tensor; trace-free tidal tensor; ablation; gravitational coherence surface; parity surface; relational signature; nested scenes; finite-source identity.

1. Motivation: gravitational scenes are nested

Gravity is rarely encountered as a single isolated source speaking into an empty universe. A moon is read inside a planet–star system. A planet is read inside a stellar system. A star is read inside a local stellar neighborhood. A galaxy is read inside a group, cluster, and cosmic-web environment. Physical sources may be finite and locally distinguishable, yet the geometry in which they participate is shared and seamless.

This nesting is obvious physically but easy to flatten conceptually. One may evaluate a field at a point, decompose a linear weak-field model into analytically assigned components, or compress a larger system into an effective mass or averaged background. These operations are lawful and often essential. Between pointwise evaluation and coarse summary, however, lies a different question:

How can the contribution of a finite source be operationally identified inside a fused gravitational scene, and where does that source-conditioned response reach parity with the context represented by the comparison scene?

The gravitational coherence surface is the simplest geometry answering the parity part of that question.

The first formulation described the GCS as a *parity horizon of gravitational identity*. That language captured the motivating intuition but invited two stronger interpretations: that the source owned a bounded parcel of geometry, and that the surface marked the end of its gravitational identity. The revised formulation makes neither claim. General relativity supplies one solved geometry for the full scene; it does not ordinarily label pieces of that geometry by source ownership. A finite source can nevertheless condition the scene in a structured and measurable way. Registered ablation reveals that source-conditioned response without pretending that it was an independently existing field component before subtraction.

The word *surface* remains appropriate. A single scalar equality imposed on a three-dimensional spatial slice generically defines a codimension-one set. At regular points, this set is a two-dimensional manifold. At critical points it may pinch, merge, self-contact, or develop seams and lower-dimensional strata. The GCS denotes the full parity set; its regular codimension-one portion is the *GCS surface stratum*.

The word *coherence* is likewise narrow. Gravity does not stop outside the GCS. An orbit does not change regime sharply merely because the surface is crossed. Coherence means only that, under one declared comparison, the source-conditioned amplitude remains larger than the ablated-scene contextual amplitude. This dependence is not a weakness. It is the point.

2. Minimal grammar of a gravitational coherence surface

The original contribution–context grammar is retained, but its preferred gravitational realization is now tensor-first and registration-aware.

Table 1: Minimal grammar of the revised GCS construction.

Element	Role in the construction
Declared scene	Source roster, geometry-construction method, physical regime, units, coordinates, domain, masks, datums, and comparison conventions.
Declared intervention	The finite source, composite source, multipole change, or roster modification whose relational signature is being tested.
Registered comparison	The full solved scene, the re-solved intervention scene, and the maps required to compare points and frames on a common support.
Witness	The tensor field or physically named proxy through which the comparison is read.
Scalarization	The Frobenius amplitude under a declared reference inner product is the default; other norms or projections are explicitly named sensitivity or proxy lanes.
GCS	The zero set on which the declared signature amplitude equals the declared contextual amplitude.

Definition 1 (Declared gravitational scene). *A declared gravitational scene specifies the physical*

source roster, geometry-construction method, regime, units, coordinates, domain, masks, datums, and comparison conventions under which a gravitational question is posed.

Definition 2 (Declared intervention). *A declared intervention identifies the source, composite source, multipole, or roster change whose source-conditioned response is being tested. Source ablation is the principal intervention considered here.*

Definition 3 (Registered comparison pair). *Let (Σ, γ_{ij}) be the full spatial scene and $(\Sigma_{-a}, \gamma_{ij}^{(-a)})$ the scene obtained after ablating source a . A registered comparison pair consists of these solved scenes together with a point identification*

$$I_a : \Sigma \supset U \longrightarrow U_{-a} \subset \Sigma_{-a}, \quad (1)$$

and a declared linear map

$$P_a(x) : T_x \Sigma \longrightarrow T_{I_a(x)} \Sigma_{-a} \quad (2)$$

used to identify or transport frames. The comparison also declares the positive-definite reference inner product used for scalar contraction.

The registration may be a gauge-fixed shared-coordinate identification, an orthonormal-frame alignment, a background-frame comparison, or another explicit protocol. Different lawful registrations need not agree exactly. Their sensitivity is part of the result rather than a nuisance to be hidden.

3. Preferred construction: registered tensor ablation

3.1 Why ablation is needed

In a linear weak-field model it is often convenient to write a total field as a sum of source-assigned fields. In nonlinear geometry, that decomposition is not generally unique or self-defining. Even in conformally constructed static initial data, the metric and its curvature mix source parameters through nonlinear functions of the combined conformal factor. The fused curvature is therefore not, in general, the sum of independently solved single-source curvatures [3, 2].

Ablation replaces assumed ownership with an operational intervention:

1. solve the full declared scene;
2. remove the declared source from the source roster;
3. solve the comparison scene under the same declared method;
4. identify points and frames between the two geometries;
5. measure the registered difference.

The intervention is counterfactual. It does not represent a source disappearing dynamically, radiating away, or being physically destroyed. It compares two solved scenes.

3.2 Electric Weyl as the preferred witness

Let n^μ be the future-directed timelike unit normal to a declared spatial slice and $C_{\mu\alpha\nu\beta}$ the Weyl tensor. The electric part of Weyl is

$$E_{\mu\nu} = C_{\mu\alpha\nu\beta} n^\alpha n^\beta, \quad (3)$$

with spatial components E_{ij} on the slice [7, 11]. It records the trace-free free-gravitational tidal structure relative to the chosen normal. In vacuum, where the Riemann tensor equals the Weyl

tensor, E_{ij} is the complete spatial tidal operator governing geodesic deviation relative to the chosen normal. Inside matter, the complete geodesic-deviation operator also contains Ricci contributions; the electric-Weyl tensor must therefore be interpreted as one declared witness rather than the whole local tidal operator.

On a time-symmetric vacuum initial slice, $K_{ij} = 0$. Under the standard constraints and sign conventions,

$$E_{ij} = {}^{(3)}R_{\langle ij \rangle}. \quad (4)$$

The vacuum Hamiltonian constraint gives ${}^{(3)}R = 0$, so this reduces to

$$E_{ij} = {}^{(3)}R_{ij}, \quad (5)$$

up to the adopted Weyl-sign convention. In the weak-field vacuum limit, E_{ij} agrees, again up to convention, with the trace-free Hessian of the Newtonian potential.

The electric-Weyl tensor is preferred because it retains both amplitude and directional structure. Its Frobenius amplitude under the declared reference inner product is the default direction-agnostic scalarization for parity extraction, but the tensor bundle remains the primary record.

3.3 Registered relational signature

The registered ablated tensor on the support of the full scene is defined by

$$\tilde{E}^{(-a)}(x) = P_a(x)^* \left[E^{(-a)}(I_a(x)) \right], \quad (6)$$

where P_a^* acts on both tensor slots. In components,

$$\tilde{E}_{ij}^{(-a)}(x) = P_i^k(x) P_j^\ell(x) E_{k\ell}^{(-a)}(I_a(x)). \quad (7)$$

The source-conditioned relational signature tensor is then

$$\Delta_a E_{ij}(x) = E_{ij}^{(\text{full})}(x) - \tilde{E}_{ij}^{(-a)}(x). \quad (8)$$

This response contains both the direct geometric change associated with removing a and the resulting change in nonlinear couplings throughout the scene. It is therefore not identified with a privately owned field of source a .

3.4 Frobenius amplitude and the ablation GCS

Let γ_{ref} be the declared positive-definite spatial reference metric. For a registered covariant tensor A_{ij} define

$$\|A\|_{F,\text{ref}} = \left(\gamma_{\text{ref}}^{ik} \gamma_{\text{ref}}^{j\ell} A_{ij} A_{k\ell} \right)^{1/2}. \quad (9)$$

In a physical orthonormal frame this becomes $\|A\|_F = (\sum_{ij} A_{ij}^2)^{1/2}$.

Define the signature amplitude

$$S_a(x) = \|\Delta_a E(x)\|_{F,\text{ref}}, \quad (10)$$

and the ablated-scene contextual amplitude

$$C_a(x) = \|\tilde{E}^{(-a)}(x)\|_{F,\text{ref}}. \quad (11)$$

The latter is the amplitude of the re-solved counterfactual scene without a ; it is not asserted to be a separable remainder already residing inside the full geometry.

Define the parity residual

$$F_a(x) = S_a(x) - C_a(x). \quad (12)$$

The ablation-defined gravitational coherence surface is

$$\Sigma_{\text{GCS}}^{(a)} = \{x \in U : F_a(x) = 0\} = \{x \in U : S_a(x) = C_a(x)\}. \quad (13)$$

The GCS does not mark where the signature vanishes. It marks where one declared amplitude of the source-conditioned response reaches equality with the geometry of the registered ablated scene.

3.5 Tensor channels beyond amplitude

The Frobenius norm is rotation-invariant within a fixed physical inner product and uses the full tensor amplitude, making it a useful default scalarization. It does not exhaust the tensor information. A registered signature bundle may also retain eigenvalues, spectral-shape ratios, principal directions, spectral norm, invariant polynomials, degeneracy proximity, angular coherence, and downstream geodesic-deviation tests.

Eigenframe rotation requires particular care. Eigenvectors are non-unique at repeated eigenvalues and unstable near spectral degeneracy. Any rotation channel should therefore be paired with an eigenvalue-gap or degeneracy-proximity diagnostic. Agreement between scalar norms is a sensitivity test; it does not make all scalarizations physically equivalent.

4. Regularity, branches, and topology

The GCS is the zero set of the parity residual F_a . If F_a is continuously differentiable and $\nabla F_a \neq 0$ at a point on the zero set, the implicit function theorem makes the GCS locally a smooth codimension-one surface. Where $F_a = 0$ and $\nabla F_a = 0$, the parity set may pinch, merge, split, self-contact, or collapse into a lower-dimensional stratum.

For a source-centered interrogation, define the registered amplitude ratio

$$\rho_a(r, \hat{n}) = \frac{S_a(r, \hat{n})}{C_a(r, \hat{n})}. \quad (14)$$

Proposition 1 (Single-branch regularity). *Suppose $\rho_a(r, \hat{n})$ is continuous in r along each declared ray, strictly monotone along that ray, and takes values both above and below unity. Then the parity condition $\rho_a = 1$ has exactly one crossing on that ray. If ρ_a is continuously differentiable and $\partial_r \rho_a \neq 0$ at the crossing, the root varies smoothly with direction.*

Proof. Existence follows from the intermediate value theorem applied to $\rho_a - 1$. Strict monotonicity gives uniqueness. Smooth dependence of the root on direction follows from the implicit function theorem because the radial derivative is nonzero at the crossing. $\square$

These conditions define a regular special case, not a universal topology. Full root inventories may contain rising and falling branches, multiple crossings, and disappearance–re-emergence sequences. An innermost-root extractor can therefore miss or misclassify the intended branch. Topology-sensitive extraction should record every crossing, orientation, confidence, and continuity across parameter steps.

Closure is a tested property of a declared comparison, not an automatic consequence of the name GCS. Four meanings must remain distinct:

- closed as a topological subset;
- compact and enclosing a declared source;
- numerically closed within a finite domain and tolerance;
- visually closed in a particular render.

Wild deformation is ordinary geometry; a true change in connectivity is a stronger claim requiring stronger tests.

5. The weak-field founding lane

The registered tensor-ablation construction is the preferred serious gravitational realization, but the founding scalar cases remain useful as transparent proxy lanes.

5.1 Acceleration-proxy GCS

For a source of mass M , a declared weak-field monopole acceleration proxy is

$$Q_M(r) = \frac{GM}{r^2}. \quad (15)$$

If a neighboring mass M' at separation D dominates the contextual acceleration scale, then

$$Q_K \approx \frac{GM'}{D^2}. \quad (16)$$

The acceleration-proxy parity radius is

$$r_0 = D\sqrt{\frac{M}{M'}}. \quad (17)$$

This construction is frame- and protocol-dependent. A nearly uniform acceleration may be removed locally by a freely falling frame, while its tidal gradient cannot. The acceleration-proxy GCS is therefore pedagogical and operational, not the unique GR-native coherence surface.

5.2 Earth–Moon–Sun as diagnostic lesson

The Moon is dynamically Earth-bound. Its motion belongs to the three-body structure of the Earth–Moon–Sun system, differential forcing, rotating-frame dynamics, and ordinary celestial mechanics. At the same time, the Sun’s absolute gravitational acceleration at the Moon is larger than Earth’s. Under the simplified acceleration-proxy rule, Earth’s monopole parity radius in the solar context is approximately

$$r_0 \approx 261,000 \text{ km}, \quad (18)$$

while the Moon’s mean orbital distance is approximately

$$r_{\text{Moon}} \approx 384,000 \text{ km}. \quad (19)$$

These statements do not conflict. At the broadest level, three questions must be separated: dynamical belonging, declared contribution–context parity, and the complete nested-scene geometry. Within the parity category, acceleration, tidal, multipole, and ablation witnesses answer different operational questions.

The Moon therefore supplies the first diagnostic lesson of the GCS program. Familiar language about being “inside Earth’s gravity” compresses several distinct relational questions.

5.3 Relation to the Hill sphere

In the circular restricted three-body approximation, the Hill radius is

$$r_H = D \left(\frac{M}{3M'} \right)^{1/3}. \quad (20)$$

The acceleration-proxy GCS and Hill scales are related by

$$\frac{r_0}{r_H} = 3^{1/3} \left(\frac{M}{M'} \right)^{1/6}. \quad (21)$$

The sixth-root dependence makes this ratio vary slowly across ordinary planetary mass ratios. This does not create a universal relationship between GCS and Hill stability; it explains why two numerically comparable scales can nevertheless encode different physical questions. The Hill sphere concerns orbital stability in a rotating approximation. The acceleration-proxy GCS concerns equality between declared acceleration magnitudes in a specified protocol. The two scales agree only on what they reject: neither is a force cutoff.

Table 2: The GCS beside neighboring influence-boundary concepts.

Boundary	Primary question	Explicit non-claim
Ablation GCS	Where does the registered tensor response to a source reach parity with the registered ablated-scene field?	Not a force wall, causal horizon, or end of the signature.
Acceleration-proxy GCS	Where do declared acceleration magnitudes match in a specified frame or protocol?	Not uniquely GR-native or frame-independent.
Hill sphere	Where can satellite orbits remain dynamically stable in a restricted three-body approximation?	Not a readability boundary.
Laplace sphere of influence	Where is patched-conic frame switching useful?	Not a tensor parity surface.
Measurement floor	What modeled signal is distinguishable under uncertainty?	A measurement layer, not the root definition.

6. Environmental dependence and multipole readability

A GCS is relational. The same source can generate different parity surfaces in different scenes because the derived or counterfactual context changes. A source near a dominant neighbor may have a compressed parity surface toward that neighbor and an elongated surface elsewhere. This geometry records anisotropic context; it does not imply a material pressure or membrane.

The original weak-field formulation illustrated this dependence across nested scales. These estimates remain useful when kept in their proper claim lane.

In the weak-field Newtonian lane, the trace-free tidal tensor is the trace-free Hessian of the potential. For an exterior multipole of order ℓ , a scalar readability amplitude may scale as

$$Q_\ell(r) \propto \frac{G|M_\ell|}{r^{\ell+3}}, \quad (22)$$

Table 3: Illustrative weak-field proxy scales inherited from the founding formulation. Values are pedagogical order-of-magnitude estimates, not ablation-GCS measurements.

Scene	Contribution tested	Scale	Interpretation
Earth in solar context	Earth monopole	$\sim 261,000$ km	Inside the Moon’s orbit; not a dynamical cutoff.
Sun in local stellar context	Solar monopole	~ 2.15 pc	Comparable to nearby-star spacing.
Jupiter in solar context	Jupiter monopole	~ 0.16 AU	A readability scale distinct from satellite stability.
Milky Way in Local Group	Galactic monopole	~ 629 kpc	Order-of-magnitude source-context scale.
Milky Way disk in Local Group	Quadrupole/disk shape	~ 81 kpc	Recoverability scale for non-spherical structure, not a torque cliff.
$10^6 M_\odot$ globular cluster	Cluster monopole	~ 36 pc / ~ 16 kpc	Same source in dense and sparse scenes; roughly $442\times$ expansion.

with angular factors and projection conventions declared separately. The mode-specific parity radius satisfies $Q_\ell(r_\ell) = Q_K$. Higher multipoles generally reach parity closer to the source than the monopole under comparable contexts.

A known contextual amplitude can constrain a readable exterior mode at parity, but it cannot uniquely reconstruct arbitrary interior matter. Many interiors share the same low-order moments [6, 8]. The ablation construction supplies a complementary route: rather than assigning one analytic mode as the contribution, it measures the full registered tensor response and studies multipole information through its angular and spectral structure.

7. Static-slice benchmark demonstration

The preferred construction has been implemented in a declared time-symmetric conformally flat initial-data lane. For Brill–Lindquist-type static data,

$$\psi = 1 + \sum_{i=1}^N \frac{m_i}{2r_i}, \quad \gamma_{ij} = \psi^4 \delta_{ij}, \quad K_{ij} = 0, \quad (23)$$

and the vacuum Hamiltonian constraint is satisfied away from punctures [3]. Because curvature is computed from the combined conformal factor, the fused electric-Weyl field is not the sum of independently solved source fields.

In the internal July 2026 benchmark, a many-source static scene was solved, one declared source was ablated, and the scene was re-solved under the same construction. The current internal implementation used an explicitly declared shared-coordinate component registration and a common Cartesian reference metric. That lane is reproducible as a diagnostic but is not claimed to be convention-free; a physical-frame registration remains the preferred public target.

The benchmark produced a global $\Delta_a E_{ij}$ field with a localized high-amplitude footprint and a compact enclosing Frobenius-parity surface. In the declared scene, the parity interior occupied approximately 1.62% of the sampled domain. Frobenius and spectral scalarizations produced closely

aligned surfaces, with a Jaccard overlap of approximately 0.93 and disagreement concentrated in a thin boundary skin. These results establish the operation in one exact static-data regime. They do not establish universal closure, norm independence, temporal persistence, or observational detectability.

A publication-grade benchmark report should include the complete source roster, coordinates, masses, masks, registration convention, reference metric, sampling resolution, root inventory, closure test, and convergence analysis. The present paper uses the benchmark only to show that the revised definition has been instantiated as a reproducible diagnostic rather than remaining a verbal proposal.

8. From relational signature to finite-source identity

The exterior ablation signature is the registered tensor field $\Delta_a E_{ij}$ and its derived channels. It is generally not a closed object and need not terminate at the GCS. The GCS is one parity level set within that global response.

The recurring word *identity* should therefore be used at the correct claim level.

Definition 4 (Finite-source relational identity hypothesis). *A finite source possesses relational gravitational identity if its source-conditioned signatures contain invariants that persist, under declared transformations, across changes of context, representation, frame, scale, and time.*

The hypothesis is stronger than the existence of an ablation signature. A signature shows that a declared intervention produces structured difference. Identity requires persistence and recognizability across multiple lawful comparisons. Evidence would include morphology discrimination, cross-context stability after declared normalization, frame and scalarization robustness, temporal tracking, and a reproducible relation between interior and exterior channels.

Failure to find any registration-, normalization-, or context-stable discriminant across controlled source families would count against the hypothesis. No sovereign territory is implied. The field is seamless; the source-conditioned response may be global; and the GCS marks parity rather than extinction.

9. Extensions: interior lodging and relational complexes

Two extensions now sit beyond the minimum founding construction.

First, finite sources may exhibit an interior lodging signature. In a weak-field spherical source, if the first nonconstant central density term has order p , then the small-context parity radius can scale as the reciprocal onset exponent, schematically $r \propto \chi^{1/p}$. This result has been derived in the Newtonian lane and observed in declared conformal-density experiments for tested profiles. Its broader physical-frame and source-class validity remains open. The detailed derivation belongs in a companion analysis rather than in the root GCS definition.

Second, a registered before/after relational complex may record which cells, faces, seams, and nodes are removed, born, or relabeled under ablation. Such a comparison is operationally meaningful only on a fixed registered datum skeleton or under an explicit matching map between complexes. The tensor bundle witnesses the geometric response; the relational complex supplies a topological reading scaffold. Neither object is a new physical wall or source.

10. Observational calibration as a measurement layer

The mathematical definition requires a derived or counterfactual context, not the word *calibrated*. Empirical use, however, requires the scene and its uncertainties to be estimated from ephemerides,

catalogs, lensing models, stellar dynamics, simulations, or instrumental reconstructions.

A measurement-aware comparison may write an effective uncertainty floor schematically as

$$Q_{\text{eff}}^2 = Q_{\text{env}}^2 + Q_{\text{inst}}^2 + Q_{\text{model}}^2. \quad (24)$$

This is a signal-to-context special case layered onto the underlying gravitational comparison. Measurement uncertainty does not create the root GCS; it determines confidence in an empirical estimate of its location and morphology.

For a tensor-ablation GCS, empirical reporting should include uncertainty in the source roster, geometry construction, registration map, frame convention, scalarization, masks, sampling, and root extraction. A clean render is not a substitute for that ledger.

11. Evidence register and non-claims

Table 4: Current claim status of the revised GCS program.

Result	Status	Scope or promotion requirement
Contribution–context parity grammar	Defined	Mathematical grammar independent of witness choice.
Weak-field acceleration and multipole cases	Implemented proxies	Pedagogical and frame/protocol dependent.
Registered electric-Weyl ablation bundle	Implemented	Exact registration and scalarization must be declared per experiment.
Compact enclosing parity surface	Reproduced internal benchmark	Tested static scenes only; vary source, scene, resolution, masks, and registration before promotion.
Frobenius versus spectral sensitivity	Reproduced internal benchmark	Agreement in one declared scene; not norm independence in general.
Interior onset law	Reproduced in tested lanes	Broader profiles and physical-frame formulation remain open.
Interior/exterior invariant relation	Seed	No cross-scale invariant yet established.
Finite-source relational identity	Hypothesis	Requires persistence across context, frame, scale, representation, and time.

The GCS does not modify general relativity, introduce a new force, or add a physical source. It is not a causal horizon, event horizon, trapping horizon, material membrane, orbital-stability boundary, or automatic observable. Ablation is a counterfactual comparison between solved scenes, not physical destruction or time evolution. $\Delta_a E_{ij}$ is not necessarily a uniquely owned field of source a . The GCS is not where the source’s gravitational effect or signature ends. A compact GCS does not imply compact support of the response. Coordinate-component subtraction is not automatically invariant. An orthonormal frame does not eliminate the need for point and frame identification between geometries. A static or adiabatic slice sequence does not establish causal evolution, radiation, or temporal persistence. Finite-source relational identity remains a hypothesis.

12. Summary

A gravitational coherence surface is a contribution–context parity surface in a declared gravitational scene. The founding formulation expressed this idea through declared readability channels. The revised formulation retains that minimal grammar while giving it a preferred operational realization:

solve the full scene, solve a registered ablated scene, measure the change in the electric-Weyl tensor, apply the declared Frobenius scalarization, and locate where the signature amplitude reaches equality with the registered ablated-scene amplitude.

This revision changes the interpretation of gravitational identity. A finite source does not own a bounded piece of the shared geometry. Its ablation response is a structured difference between two solved scenes, including altered nonlinear couplings. The response may extend throughout the modeled domain. The GCS is one codimension-one parity level set within it, not its outer edge.

The weak-field founding examples remain useful. Earth–Moon–Sun acceleration parity differs from Hill stability because the constructions ask different questions. Environmental dependence remains central because the GCS belongs to a source-in-a-scene rather than to a source alone. Multipole and observational lanes remain valid when their witnesses and uncertainties are declared.

The program now distinguishes signature from identity. Registered ablation establishes a source-conditioned relational signature. Whether invariant features of that signature persist strongly enough across context, frame, scale, representation, and time to deserve the name *identity* is a precise research hypothesis.

Finite matter does not carry sovereign gravitational territory. It may nevertheless condition the shared geometry in a persistent and source-readable way. The GCS is not where that signature ends. It is where the default Frobenius measure of the registered signature reaches parity with the registered ablated-scene context under the declared comparison convention.

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